2024 AMC 10B Problem 20 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 20

Problem:

Three different pairs of shoes are placed in a row so that no left shoe is next to a right shoe from a different pair. In how many ways can these six shoes be lined up?

Answer Choices:

A. 60

B. 72

C. 90

D. 108

E. 120

Solution:

There are $\binom{6}{3} = 20$ arrangements of the letters LLLRRR representing the positions of 3 left shoes and 3 right shoes in the row of 6 shoes. Of the 20 , any sequence containing LRL or RLR will violate the condition given in the problem. There are 8 arrangements that avoid these two sequences. Call a pair of shoes matched if the 2 shoes in the pair are next to each other. There are three sets of possibilities.

- LLLRRR and RRRLLL: In these two cases, only one pair of shoes is matched. There are 3 choices for that pair, and there are $2 \cdot 2 = 4$ ways to place the other 4 shoes for a total of $6 \cdot 4 = 24$ arrangements for this case.

- LRRRLL, LLRRRL, RLLLRR, and RRLLLR: In each of these four cases, there are $3! = 6$ ways to place the left shoes, but then there is a unique way to place the right shoes, for a total of $6 \cdot 4 = 24$ arrangements in this case.
- LRRLLR and RLLRRL: In these two cases, all three pairs of shoes are matched, so, in each case, there are $3! = 6$ ways to place the shoes. This gives a total of $6 \cdot 2 = 12$ arrangements.

Thus there are $24 + 24 + 12 = \text{(A)} \boxed{60}$ arrangements satisfying the conditions of the problem.

OR

Label the shoes L_i and R_i for $i = 1, 2, 3$. By symmetry it suffices to count the number of arrangements with L_1 first in line and multiply by 6 . There are 2 choices for a left shoe coming next, say
L_2, after which the only continuations are $L_3 R_3 R_j R_{3-j}$ for $j = 1$ or 2 , or $R_2 R_1 R_3 L_3$; this gives $2 \cdot (2 + 1) = 6$ arrangements. Otherwise R_1 comes second, followed by either $R_k R_{5-k} L_{5-k} L_k$ or $R_k L_k L_{5-k} R_{5-k}$ for $k = 2$ or 3 , another $2 + 2 = 4$ arrangements. This gives a total of 10 , so there are $6 \cdot 10 = \text{(A)} \boxed{60}$ ways to line up the six shoes.

Note: Generalized to n pairs of shoes, the sequence of answers to this question starts $2, 8, 60, 816$, 17520. This is sequence A096121 in the On-Line Encyclopedia of Integer Sequences, which is listed as counting the number of paths of a rook on a $2 \times n$ board, where the rook must land on each square exactly once. The sequence satisfies the recurrence $a_{n+1} = n(n+1)\left(a_n + a_{n-1}\right)$.

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